

LAB 11

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BASH Scripting

1. Write a bash script (filecheck.sh) that checks whether a regular file exists or not. The script must be able to check itself and other files as an argument of the script. Show the script's code and run the script as below:

> sudo nano filecheck.sh

```

GNU nano 4.8                                filecheck.sh                                Modified
# Check if a file is provided as an argument
if [ -z "$1" ]; then
    echo "Usage: $0 <filename>"
    echo "Checking the script itself..."
    if [ -f "$0" ]; then
        echo "The script '$0' exists."
    else
        echo "The script '$0' does not exist."
    fi
else
    if [ -f "$1" ]; then
        echo "The file '$1' exists."
    else
        echo "The file '$1' does not exist."
    fi
fi

```

> sudo chmod 755 filecheck.sh

- a) Run the script without any argument. What is the output? Explain the output.

```

osboxes@osboxes:~$ ./filecheck.sh
Usage: ./filecheck.sh <filename>
Checking the script itself...
The script './filecheck.sh' exists.

```

- b) Run the script with **lab11.txt** as argument. What is the output? Explain the output.

```

osboxes@osboxes:~$ ./filecheck.sh lab11.txt
The file 'lab11.txt' does not exist.

```

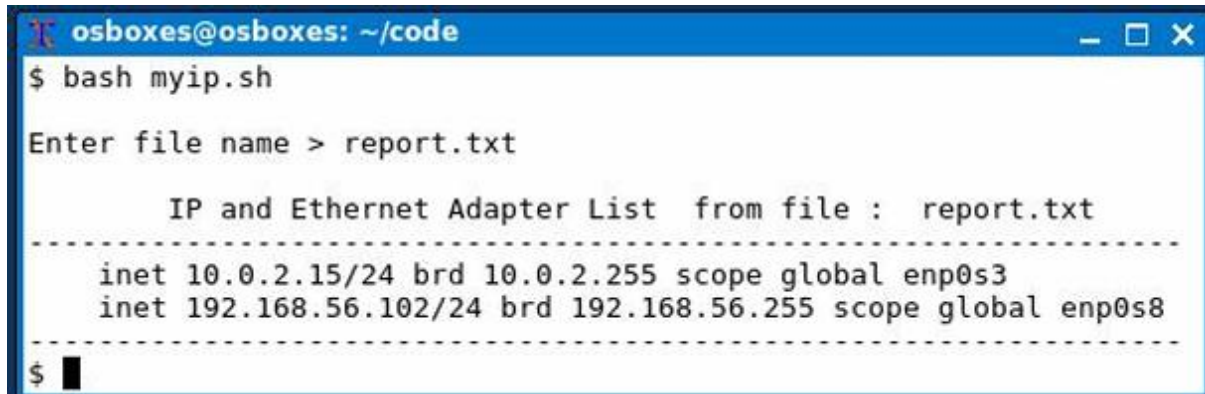
Script check for the file with the name, if it does not exist in the system, it will echo the file does not exist.

c) Create a file **lab11.txt** using touch. Run the script again as Q1b.

> touch lab11.txt

```
osboxes@osboxes:~$ touch lab11.txt
osboxes@osboxes:~$ ./filecheck.sh lab11.txt
The file 'lab11.txt' exists.
```

2. Write a bash script called “**myip.sh**” that shows your computer/server’s IP address, broadcast address and Ethernet interface. The script also prompts user to key-in file name to save all the content. Possible output is as below:

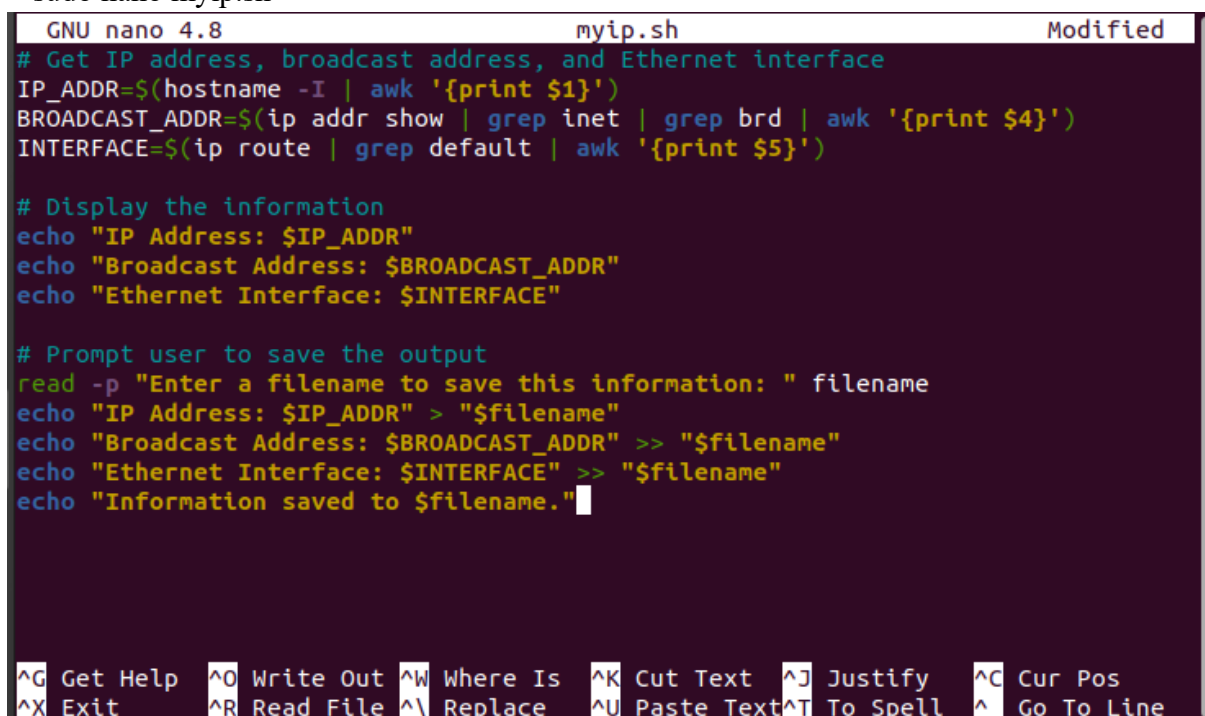


```
osboxes@osboxes: ~/code
$ bash myip.sh

Enter file name > report.txt

      IP and Ethernet Adapter List  from file :  report.txt
-----
      inet 10.0.2.15/24 brd 10.0.2.255 scope global enp0s3
      inet 192.168.56.102/24 brd 192.168.56.255 scope global enp0s8
-----
$
```

> sudo nano myip.sh



```
GNU nano 4.8          myip.sh          Modified
# Get IP address, broadcast address, and Ethernet interface
IP_ADDR=$(hostname -I | awk '{print $1}')
BROADCAST_ADDR=$(ip addr show | grep inet | grep brd | awk '{print $4}')
INTERFACE=$(ip route | grep default | awk '{print $5}')

# Display the information
echo "IP Address: $IP_ADDR"
echo "Broadcast Address: $BROADCAST_ADDR"
echo "Ethernet Interface: $INTERFACE"

# Prompt user to save the output
read -p "Enter a filename to save this information: " filename
echo "IP Address: $IP_ADDR" > "$filename"
echo "Broadcast Address: $BROADCAST_ADDR" >> "$filename"
echo "Ethernet Interface: $INTERFACE" >> "$filename"
echo "Information saved to $filename."

^G Get Help  ^O Write Out ^W Where Is  ^K Cut Text  ^J Justify   ^C Cur Pos
^X Exit      ^R Read File ^\ Replace   ^U Paste Text ^T To Spell  ^_ Go To Line
```

> sudo chmod 755 myip.sh

> ./myip.sh

```

osboxes@osboxes:~$ ./myip.sh
IP Address: 192.168.248.207
Broadcast Address: 192.168.248.255
Ethernet Interface: enp0s3
Enter a filename to save this information: report.txt
Information saved to report.txt.
osboxes@osboxes:~$ bash myip.sh
IP Address: 192.168.248.207
Broadcast Address: 192.168.248.255
Ethernet Interface: enp0s3
Enter a filename to save this information: report1.txt
Information saved to report1.txt.

```

OR

```

> sudo nano myip1.sh
> sudo chmod 755 myip1.sh
> ./myip1.sh

```

```

# Prompt user to enter a file name
read -p "Enter file name > " filename

# Get IP and Ethernet adapter information
ip_info=$(ip addr show | grep inet | grep brd)

# Save the information to the specified file
echo "$ip_info" > "$filename"

# Display the output in the desired format
echo -e "\n          IP and Ethernet Adapter List from file : $filename"
echo "- - - - -"
echo "$ip_info" | sed 's/^/ /'
echo "- - - - -"

osboxes@osboxes:~$ ./myip1.sh
Enter file name > Report3.txt

          IP and Ethernet Adapter List from file : Report3.txt
- - - - -
      inet 192.168.248.207/24 brd 192.168.248.255 scope global dynamic enp0s3
- - - - -

```

3. Write a bash script to calculate TWO integers and print that on the screen. Use 'let' to perform multiplication of number 10 and 3. Give 'calculate.sh' as script's name.

> sudo nano calculate.sh

```
GNU nano 4.8 calculate.sh Modified
# Perform multiplication using 'let'
let result=10*3
echo "The result of 10 * 3 is: $result"
```

> sudo chmod 755 calculate.sh

> ./calculate.sh

```
osboxes@osboxes:~$ sudo nano calculate.sh
osboxes@osboxes:~$ sudo chmod 755 calculate.sh
osboxes@osboxes:~$ ./calculate.sh
The result of 10 * 3 is: 30
```

4. You can do the arithmetical operations in bash in different ways such as through 'let' and using 'double bracket'. Write a script that adds two integer numbers using double brackets. Two integer values will be taken from the user's input and print the result of addition. Save the script's file as 'add_numbers.sh'.

> sudo nano add_numbers.sh

```
GNU nano 4.8 add_numbers.sh Mo
# Prompt user for two integers
read -p "Enter the first integer: " num1
read -p "Enter the second integer: " num2

# Perform addition using double brackets
result=$((num1 + num2))

# Display the result
echo "The result of $num1 + $num2 is: $result"
```

> sudo chmod 755 add_numbers.sh

> ./add_numbers.sh

```
osboxes@osboxes:~$ sudo chmod 755 add_numbers.sh
osboxes@osboxes:~$ ./add_numbers.sh
Enter the first integer: 10
Enter the second integer: 3
The result of 10 + 3 is: 13
```

Useful Sysadmin Commands

5. Identify and run THREE useful Sysadmin commands as below:

a) **List block devices.** Identify how many partitions in your system and their sizes.

> lsblk

```
osboxes@osboxes:~$ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINT
loop0       7:0      0   61.9M  1 loop /snap/core20/1328
loop1       7:1      0   63.7M  1 loop /snap/core20/2434
loop2       7:2      0    1.5G  1 loop /snap/root-framework/936
loop3       7:3      0   73.9M  1 loop /snap/core22/1722
loop4       7:4      0   44.3M  1 loop /snap/snapd/23258
loop5       7:5      0   73.9M  1 loop /snap/core22/1663
loop6       7:6      0   66.2M  1 loop /snap/core24/609
loop7       7:7      0   67.2M  1 loop /snap/lxd/21835
loop8       7:8      0   91.9M  1 loop /snap/lxd/29619
loop9       7:9      0   44.4M  1 loop /snap/snapd/23545
loop10      7:10     0    1.6G  1 loop /snap/root-framework/943
loop11      7:11     0   66.2M  1 loop /snap/core24/716
sda         8:0      0   500G   0 disk
├─sda1      8:1      0     1M   0 part
├─sda2      8:2      0   236G   0 part /
├─sda3      8:3      0   300M   0 part /boot
├─sda4      8:4      0    9.5G   0 part [SWAP]
└─sda5      8:5      0  254.2G   0 part /home
sr0        11:0     1     57M   0 rom  /media/osboxes/VBox_GAs_7.1.4
```

b) **Estimate file space usage.** Summarize disk usage of /var folder for only one level with human readable output.

> sudo du -sh /var

```
osboxes@osboxes:~$ sudo du -sh /var
5.4G    /var
```

[-h] is human-readable format

[-s] ensures only a summary is shown

```
osboxes@osboxes:~$ sudo du -s /var
5631728 /var
osboxes@osboxes:~$ sudo du -h /var
4.0K    /var/snap/lxd/21835
4.0K    /var/snap/lxd/common/lxd
12K     /var/snap/lxd/common
4.0K    /var/snap/lxd/29619
24K     /var/snap/lxd
4.0K    /var/snap/core24/609
4.0K    /var/snap/core24/common
4.0K    /var/snap/core24/716
16K     /var/snap/core24
4.0K    /var/snap/core22/1722
4.0K    /var/snap/core22/common
4.0K    /var/snap/core22/1663
16K     /var/snap/core22
4.0K    /var/snap/root-framework/943
4.0K    /var/snap/root-framework/936
```

c) Use **kill** command. Run **nano** in background. Use **ps** to find the process id. Kill the process.

> nano &

> ps aux | grep nano

> [kill <process_id>]

> kill -9 5246

[-9] force kill

```
osboxes@osboxes:~$ ps aux | grep nano
osboxes      5246  0.0  0.0   6500  3256 pts/0    T   02:43   0:00 nano
osboxes      5255  0.0  0.0   6632  3240 pts/0    T   02:44   0:00 nano
osboxes     16759  0.0  0.0   6300    720 pts/0    S+  03:22   0:00 grep --color=auto nano
osboxes@osboxes:~$ kill -9 5246
osboxes@osboxes:~$ ps aux | grep nano
osboxes      5255  0.0  0.0   6632  3240 pts/0    T   02:44   0:00 nano
osboxes     16762  0.0  0.0   6300    720 pts/0    S+  03:22   0:00 grep --color=auto nano
[1]-  Killed                  nano
osboxes@osboxes:~$
```